Forward Test: Request for Approval Before Live Paper Trading

FYP: Learned Market Making on Kalshi Event Contracts
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Purpose. Before I officially start live paper trading, I would like your approval on two things: the **live paper-trading setup** (Section 2) and the **baselines** it is compared against (Section 3). Sections 4–6 summarise the evidence behind the headline model and the decision not to tune it. The specific questions are in Section 7.

1 Strategy and headline model

The strategy is a passive market maker on Kalshi's multi-outcome markets. Every hour it considers posting a buy or sell limit order (10 contracts) on each leg, and a learned model decides which orders to post: an order is posted only if its predicted net profit after fees exceeds 1 c per contract. The headline model, **G2**, is a gradient-boosted tree model on 14 market features (price, spread, recent trades, basket structure, fees, category), retrained on all data up to August 2026. Its predecessor **G** (trained on Nov 2024–May 2025) passed a pre-registered test on Aug–Nov 2025 (Section 4). G2 was trained on 23 market series; the live test covers every series the strategy quotes (290 non-weather series plus the daily temperature series). Weather markets were first excluded on one validation month, then re-included after a pre-registered check over six walk-forward months (weather +1.60% per month on average, positive in 5 of 6; other markets +4.94%). The model's edge on weather declined in August–September (net per contract on its picks +1.96 c in July, +0.28 c in August, −0.99 c in September, while the market itself was no harder), so weather and non-weather results are also reported separately. All rules, models and statistics were fixed and timestamped publicly before any live result.

2 Live paper-trading setup

No knowledge of the future at any step. This is live paper trading, not a replay: the forward-test result is the ledger as recorded hour by hour during the test, never a recomputation after the fact.

- **Models** were trained and frozen before live trading (their hashes are public); nothing is retrained or adjusted during the test.
- **Decisions** for each hour use only data available at the live market snapshot taken at the start of that hour (about two minutes in); only trades *after* the snapshot can fill the hour's orders. Each decision is timestamped when made, and results are also reported counting only trades after that timestamp.
- **Fills** for each hour are determined only from Kalshi trades that occurred *during* that hour, after it has ended; no later trade, price or outcome is used.
- **Profit** becomes realised only when a market actually settles; until then, open positions are valued at the latest recorded market mid-price.
- **Every record is timestamped** on Lumid Research Fleet shortly after its hour, contains only that hour's data, and is never revised afterwards, so no result can be adjusted with hindsight.

1. **Live decisions.** A producer on a Lumid sandbox scores every candidate order each hour from Kalshi's live data and records every candidate's features and the main models' scores before outcomes are known; the baseline rules and models are applied to these same logged features. Each hour's record is posted to Lumid Research Fleet with a SHA-256 hash, giving an independent timestamp.
2. **Live fills, in hourly batches.** The strategy itself is hourly: it posts its orders once per hour, and they rest until the end of that hour. Fills are therefore processed in the same hourly batches. Shortly after each trading hour ends (typically within 1–2 hours), a paper trader determines that hour's fills *only from the Kalshi trades that occurred during it*, records them and timestamps them; they are never revised. Because an hour's fills depend only on that hour's trades, processing them trade by trade would give the same fills, apart from rare cases where a position limit is reached within the hour. The fill rules are the same as the backtest's:
    - an order joining the best price fills only if real trades go *through* its price (others are ahead in the queue); an order that improves the price by 1 c fills on trades at its price;
    - position limits (50 contracts per leg, 200 per event), Kalshi maker fees, full collateral;
    - positions are held to settlement and valued at the actual results.
3. **Append-only ledger.** Each hour's fills are written to a ledger entry that is never revised. Entries are hash-chained (each contains the previous entry's hash). Every round re-checks the recorded fills of markets active in the last four days against a recomputation; any mismatch is logged, never overwritten. Settlement profit is added as markets resolve.
4. **Hourly reporting.** Each hour, for every strategy, Research Fleet receives the *cumulative* realised P&L (settled markets), unrealised P&L (open positions at the latest recorded mid), total P&L, capital committed and return (pessimistic fills; the other fill assumptions are kept in the ledger), plus the ledger entry's hash. Hourly P&L is the change in total P&L.
5. **Fill assumptions.** Pessimistic fills (above) decide the verdicts; optimistic fills (trades at our price also count) and strict fills (competitors match our price improvements) are reported as bounds.
6. **What counts.** Only hours recorded live, from the first ledger entry onwards, count towards the forward-test result. The hours before the start (11–14 October), when decisions were live but fills were not yet recorded live, are shown separately for reference only and do not count. The interim cut for the report is 28 October; quoting ends 11 November.
7. **Verdicts (pre-registered, one-sided 5%).** (a) G2 is profitable live: the lower 95% bound of its net return is above zero; (b) G2 adds value: G2 minus random selection at the same volume is above zero. Confidence intervals are bootstrapped by event (and by day, as a sensitivity check).

Checks done. The paper trader reproduced exactly the 1.1 million orders the producer logged (0 mismatches); two blind test rounds in the production sandbox (no P&L viewed) confirmed the ledger, the hash chain, re-verification and an unchanged producer.

3 Baselines

All baselines run on the same hours, candidates, fill rules and statistics as G2.

Baseline	Rule	Question it answers
Random selection	random orders at G2's posting volume	Is profit from choosing well, or from trading less?
Quote everything	every candidate order	Does any selection help at all?
Wide spreads	post if spread ≥ 3 c	Is G2 just a spread filter?
Liquid markets	post if recent trades $\geq$ hourly median	Is G2 just a liquidity filter?
Linear model	ridge regression, same data and features	Is a nonlinear model needed?
Neural network	2-layer MLP, same data and features	Does another model family do better?
Previous model (G)	2025 model, unchanged	Has the edge decayed?
S&P 500, 3-month T-bill	index and risk-free rate	Better than passive investment?

Secondary variants (reported, never verdicts): G2 trained on shorter windows, a two-price-level version, and a weekly category-selection variant.

4 Evidence before the live test

2025 test (Aug–Nov 2025, pre-registered; full simulator, pessimistic fills).

Strategy	Return	95% CI	Fills	Net P&L
G (headline 2025)	**+3.55%**	[+2.51, +4.60]	34,086	+$3,748
Linear model	+4.03%	[+2.94, +5.14]	18,305	+$2,427
Neural network	+2.11%	[+1.30, +2.92]	43,975	+$3,104
Wide spreads	+0.07%	[−0.45, +0.55]	97,517	+$206
Liquid markets	−0.05%	[−0.51, +0.39]	99,930	−$145
Random (5 seeds, mean)	−0.23%	each includes 0	≈24,400	
Quote everything	−0.34%	[−0.77, +0.10]	110,867	−$1,133

G minus each baseline (paired by event): random +3.2 to +4.4 (all five 95% CIs exclude 0); wide spreads +3.47 [+2.50, +4.45]; liquid +3.60 [+2.64, +4.55]; neural +1.43 [+0.47, +2.38]; linear −0.48 [−1.73, +0.81] (no significant difference). Under strict fills, G's edge falls to +0.39% [−0.96, +1.78].

2026 walk-forward (Apr–Sep 2026, each month trained only on earlier months; offline scoring, pessimistic fills). Return on capital per month (%); net P&L summed over the six months:

Strategy	Apr	May	Jun	Jul	Aug	Sep	Mean	Pos. months	Net P&L
G2 (headline)	**+1.95**	**+2.30**	**+3.26**	**+3.46**	**+2.59**	**+2.36**	**+2.65**	**6/6**	**+$3,291**
Linear model	+3.13	+5.59	+3.73	+2.84	+3.76	+3.49	+3.76	6/6	+$1,455
Neural network	+2.83	+3.81	−0.56	+1.47	−0.93	−0.63	+1.00	3/6	
Random (same volume)	−0.66	−0.72	−0.48	−1.30	−1.72	−2.04	−1.15	0/6	−$924
Wide spreads	−0.23	−0.56	−0.25	−0.52	−2.01	−1.72	−0.88	0/6	
Liquid markets	−0.60	−0.94	−0.47	−1.53	−1.51	−0.76	−0.97	0/6	
Quote everything	−0.73	−0.79	−0.26	−1.07	−1.67	−1.45	−0.99	0/6	−$4,495

G2 made money in every month; random selection, both rules and quoting everything lost money in every month. The linear model's higher return comes with about a third of G2's capital and less than half its profit (Section 5). Offline scoring leaves out the simulator's inventory limits, so live returns are expected to be lower: on September 2026 in the full simulator, G2 made +1.41% [−1.47, +4.31] (offline +2.36%).

5 Why G2 stays the headline model (and why not linear model)

The linear model shows a higher return on capital because it trades less: in the 2025 test it posted 13.5% of candidate orders against 19.7% for G, and had about half as many fills (18,305 vs 34,086). G earns more total profit (2025: +$3,748 vs +$2,427; 2026: more profit in every month), and their returns do not differ significantly. When both models post the *same* number of orders, G2 earns more than both the linear model and the neural network at every volume tested in 2026. In 2025, G leads at the volume it actually trades (about 20% of candidates: +5.30% vs +4.59% linear and +4.36% neural); the models converge at very high volumes, which I interpret as the easier 2025 market rather than overlapping picks (overlap rises similarly in both years). G2 is therefore kept as the headline model.

6 Hyperparameters: fixed, not tuned

Decision. Model settings were fixed in advance and written into the pre-registration before any test. Instead of tuning them, I measured how much tuning could gain, using the same procedure tuning would use: each setting is trained only on earlier months and scored on the next month (walk-forward), over several months.

Boosted model (G2). Fourteen settings: G2's default and 13 variants, each changing one parameter. Return on capital per test month (offline scoring, pessimistic fills):

Setting	Apr	May	Jun	Jul	Aug	Mean	Worst
Leaves 15 (best)	+2.61	+3.08	+2.71	+3.60	+2.56	+2.91	+2.56
L2 regularisation 10	+2.52	+2.36	+2.61	+3.87	+2.32	+2.73	+2.32
G2 default	**+1.95**	**+2.30**	**+3.26**	**+3.46**	**+2.59**	**+2.71**	**+1.95**
Feature sampling 0.8	+2.84	+2.46	+2.56	+3.46	+1.91	+2.65	+1.91
Min. leaf 1000	+2.41	+2.56	+2.02	+3.71	+1.37	+2.41	+1.37
Min. leaf 100	+1.85	+3.63	+1.85	+3.51	+0.93	+2.35	+0.93
800 trees	+2.18	+1.11	+3.26	+3.50	+1.72	+2.35	+1.11
Early stop on last month	+3.73	+2.65	+2.66	+1.44	+0.93	+2.28	+0.93
Leaves 63	+2.16	+2.92	+2.27	+2.09	+1.66	+2.22	+1.66
Recency weight (3 months)	+1.61	+3.41	+2.68	+1.27	+2.11	+2.22	+1.27
Bagging (5 models)	+1.81	+2.26	+2.96	+2.35	+1.50	+2.18	+1.50
Learning rate 0.1, 120 trees	+2.70	+2.05	+2.53	+2.66	+0.90	+2.17	+0.90
Recency weight (6 months)	+0.78	+2.95	+1.63	+1.36	+0.99	+1.54	+0.78
Absolute-error loss	−0.51	−0.07	+0.48	−0.47	−0.29	−0.17	−0.51
Quote everything (reference)	−0.73	−0.79	−0.26	−1.07	−1.67	−0.90	−1.67

All values in %. The default ranks third of fourteen. Every variant using the same (squared-error) loss is profitable in every month; only the change of loss function removes the edge.

Neural network (baseline). Eight settings: the default and 7 variants, one change each, on the same months plus September 2026; G2 shown for reference:

Setting	Apr	May	Jun	Jul	Aug	Sep	Mean	Worst	Months >0
Larger (128, 64)	+2.04	+3.69	+0.38	+3.50	−0.77	−0.16	+1.45	−0.77	4/6
L2 1e-2	+1.98	+5.20	−0.20	+1.30	−0.57	−0.48	+1.21	−0.57	3/6
Deeper (64, 32, 16)	+3.78	+4.53	−2.03	+1.31	+0.42	−1.27	+1.12	−2.03	4/6
L2 1e-3	+1.97	+4.36	+0.33	+1.44	−0.82	−0.62	+1.11	−0.82	4/6
Default (64, 32)	+2.83	+3.81	−0.56	+1.47	−0.93	−0.38	+1.04	−0.93	3/6
tanh activation	+2.61	+4.97	−1.90	+0.36	−1.45	−0.13	+0.74	−1.90	3/6
Learning rate 3e-4	+1.32	+3.77	−0.42	+1.87	−0.75	−1.56	+0.71	−1.56	3/6
Smaller (32)	+1.26	+2.58	−0.31	+1.81	−1.00	−1.49	+0.47	−1.49	3/6
G2 (reference)	**+1.95**	**+2.30**	**+3.26**	**+3.46**	**+2.59**	**+2.36**	**+2.65**	**+1.95**	**6/6**

Why the settings were not tuned.

1. **The possible gain is small.** The best of the fourteen boosted settings beat the default by 0.2 percentage points per month (+2.91% vs +2.71%), much less than G2's own month-to-month variation (+1.95% to +3.46%). A difference this small cannot be told apart from noise over five months.

2. **The best setting's lead is partly luck.** Choosing the top scorer among many settings inflates its apparent advantage (Cawley & Talbot, 2010); on new data the expected gain is smaller still.

3. **Tuning carries overfitting risk in finance.** Each extra search on short, noisy, changing financial data is another chance to fit noise (Bailey et al., 2017; Arnott, Harvey & Markowitz, 2019). With a negligible expected gain, fixed settings are the safer choice; where tunability is low, defaults are appropriate (Probst, Boulesteix & Bischl, 2019).

4. **Tuning the neural network could not change the comparison.** Its best setting is about 1.2 points behind G2 and still loses money in two of six months; tuning chooses among such settings. On the broader 2025 data the default network does better (+2.11%) but G remains significantly ahead (+1.43, CI [+0.47, +2.38]).

5. **Fixed settings keep the tests credible.** Settings were recorded before any test; changing them after seeing results would weaken the pre-registered tests for almost no expected gain.

Reporting results across all reasonable settings, rather than a single tuned best, follows the logic of specification-curve analysis (Simonsohn, Simmons & Nelson, 2020). **Limitation:** settings were varied one at a time, so combinations were not searched; a joint search is left as future work. The training window was chosen by a six-month walk-forward (all history +2.65% per month vs 3-month +1.53% and 6-month +1.88%).

7 Questions for your approval

☐ Does the live paper-trading setup (fills recorded in hourly batches, matching the strategy's hourly quotes, from trades that have already happened; append-only ledger; hourly Research Fleet reports) meet the live paper-trading requirement?

☐ Are the baselines in Section 3 sufficient?

☐ Is it acceptable to keep the fixed model settings for G, G2 and neural network, given the sensitivity checks in Section 6 which justifies that there is no need to tune the model?

☐ Is G2 as the headline model acceptable, given that the linear model has a higher return but lower total profit?

☐ Is anything else you would like reported each hour or at the cuts (28 October, 11 November)?

If you approve, I will start the live ledger immediately; every hour from that point counts towards the result.